

Volume of a rectangular prism

A rectangular prism has six flat sides and all of its faces are rectangles. Multiply the length by the width by the height to find the volume of a rectangular prism.

formula also written
 $v = l \times w \times h$, or $v = lwh$

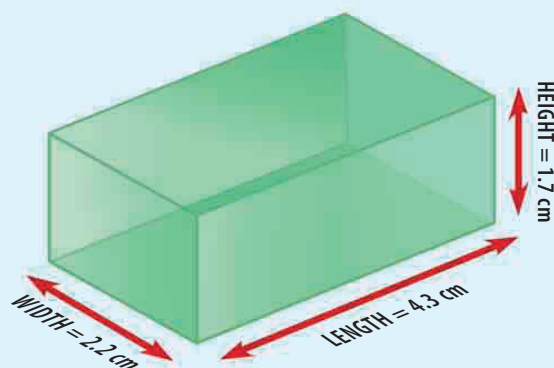
$$\text{volume} = \text{length} \times \text{width} \times \text{height}$$

$$4.3 \times 2.2 \times 1.7 = 16 \text{ cm}^3$$

▷ Multiply lengths of the sides

This rectangular prism has a length of 4.3 cm, a width of 2.2 cm, and a height of 1.7 cm. Multiply these measurements to find its volume.

answer rounded to 2 significant figures



Finding the volume of a cone

Multiply the distance from the tip of the cone to the center of its base (the vertical height) with the area of its base (the area of a circle), then multiply by $\frac{1}{3}$.

also called the
perpendicular height

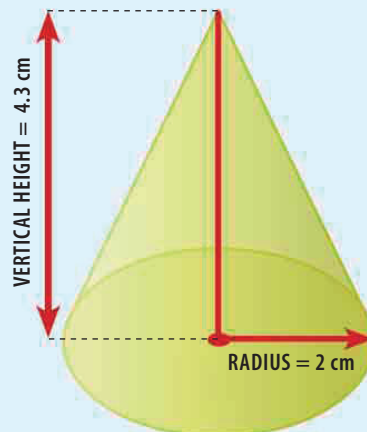
$$\text{volume} = \frac{1}{3} \times \pi \times r^2 \times \text{vertical height}$$

$$\frac{1}{3} \times 3.14 \times 2 \times 2 \times 4.3 = 18 \text{ cm}^3$$

▷ Using the formula

To find the volume of this cone, multiply together $\frac{1}{3}$, π , the radius squared, and the vertical height.

answer rounded to 2 significant figures



Finding the volume of a sphere

The radius is the only measurement needed to find the volume of a sphere. This sphere has a radius of 2.5 cm.

$$\text{volume} = \frac{4}{3} \times \pi \times r^3$$

multiply
radius by
itself twice

$$\frac{4}{3} \times 3.14 \times 2.5 \times 2.5 \times 2.5 = 65 \text{ cm}^3$$

▷ Using the formula

To find the volume of this sphere, multiply together $\frac{4}{3}$, π , and the radius cubed (the radius multiplied by itself twice).

answer rounded to 2 significant figures

